

Open source projects in NFV

Open source projects are playing a key role in the evolution and adoption of cloud technologies in general, and NFV in particular. These projects are also well supported by the industry.

In this section, we will look at some well-known open source projects relating to NFV. A brief about each project is provided along with a mention as to where they fit in the ETSI NFV reference architecture.

There is a rich open source ecosystem covering various aspects of virtualisation, SDN, NFV and the cloud. Thus, this section is not and cannot be exhaustive. Nor is it intended to be an endorsement or otherwise of any specific project mentioned or omitted here.

QEMU/KVM (wiki.qemu.org, www.linux-kvm.org):

KVM, together with QEMU, is a basic hypervisor providing both processor and board virtualisation. It is also integrated into the Linux kernel and hence is widely available and increasingly used in many cloud deployments.

In the ETSI reference architecture, QEMU/KVM provides the virtual compute capability.

Data Plane Development Kit - DPDK (www.dpdk.org): The data plane deals with handling packets/frames and applying the required functionality. The performance of networking equipment depends on the performance of the data plane. To build a high performance data plane, network equipment vendors have been using custom ASICs or network processors (NPU's). The NPUs are specialised processors optimised for packet processing.

However, with advances in CPU architecture and the advent of multi-core x86 processors, it is possible to use DPDK to create reasonably high-performance data planes running on x86 processors. DPDK provides a set of libraries, drivers and template applications to help create a data plane application.

In the ETSI reference architecture, DPDK is a building block to create VNFs.

Open vSwitch (openvswitch.org): The Open vSwitch project provides an open source virtual Ethernet switch. VNFs can comprise multiple VMs that use a private Ethernet network to interconnect them. This would be the case when a legacy chassis based network application is being virtualised.

Open vSwitch can also be used to connect multiple VNFs into a single virtual Ethernet switch, thereby providing the ability to locate VNFs in different physical nodes. It provides a virtual LAN overlay over the physical Ethernet network of a data centre.

In the ETSI reference architecture, Open vSwitch comes under the virtual network component. It can also be considered as a VNF providing switching functionality.

OpenDaylight (www.opendaylight.org): OpenDaylight is an open source SDN controller backed by some of the biggest names in the networking industry. It is also a project that comes under the Linux Foundation. According to the website, "OpenDaylight (ODL) is a highly available, modular, extensible, scalable and multi-protocol controller infrastructure built for SDN deployments on modern heterogeneous multi-vendor networks. It provides a model-driven service abstraction

platform that allows users to write apps that easily work across a wide variety of hardware and southbound protocols."

In the context of NFV, OpenDaylight can be used together with compatible Ethernet switches to create an SDN network for service chaining.

In the ETSI reference architecture, OpenDaylight primarily covers the virtualised infrastructure manager for the network functionality.

ONOS (onosproject.org): ONOS describes itself as "... the first open source SDN network operating system." ONOS is promoted by both service providers and networking vendors, and aims to provide a network operating system that caters to mission critical needs such as high-availability, scale and performance. On the southbound interface it supports OpenFlow protocol. On the northbound interface it provides abstractions for network information and APIs, using which applications can communicate 'intent' and let ONOS take care of the details, such as how to provision the various devices to implement the intent.

In the ETSI reference architecture, ONOS covers the virtualised infrastructure manager for network functionality.

OpenStack (www.openstack.org): Among the important open source projects in the area of NFV (and for cloud computing, in general) is OpenStack, which describes itself as a cloud operating system. OpenStack helps manage physical compute, storage and networking resources, and provides APIs to create, connect and manage virtual resources. It also provides an API to help create orchestration applications.

In the ETSI reference architecture, OpenStack provides the virtualised infrastructure manager along with basic MANO capabilities that can be used as building blocks to create cloud management applications.

OPNFV (www.opnfv.org): The Open Platform for NFV (OPNFV) is an initiative to create an integrated carrier-grade and open platform for NFV. It is also a project that comes under the Linux Foundation. The project aims to work with many upstream open source projects (such as OpenStack, OpenDaylight, Open vSwitch, etc), and integrate and test them for various use cases. What the industry gets is an integrated and tested platform which can become a base for NFV deployments, thereby enabling faster adoption.

OPNFV brings together various independent open source technologies, and provides a complete platform that covers the various architectural blocks shown in ETSI's reference architecture. 

References

- [1] ETSI NFV ISG: <http://www.etsi.org/technologies-clusters/technologies/nfv>
- [2] AT&T Domain 2.0 Whitepaper: https://www.att.com/Common/about_us/pdf/AT&T%20Domain%202.0%20Vision%20White%20Paper.pdf

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